

Energy Infrastructure Futures: A Multiscale Evaluation of Projected Power Plant Siting Across the Western Interconnection

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Key Points:

- Alternative future pathways lead to diverging power plant landscapes across the Western US
- Regions of the Western US are projected to see development regardless of future socioeconomics and atmospheric conditions
- Grid interconnection cost is the dominant driver of siting locations for most technologies

Abstract

The US Western Interconnection is facing unprecedented challenges in the form of less predictable peak energy demand, increasingly diverse generating resources, and fast-growing loads due to the onset of artificial intelligence, hyperscale computing, and electrification. Projecting where future generation may be developed is critical to maintaining a robust and resilient electric grid under this mounting uncertainty and variability. Using an integrated multisectoral, multiscale modeling framework that links a human-Earth systems model, an hourly load model, a geospatial power plant siting model, and an hourly grid operations model, we evaluate the power plant landscape evolution under eight alternative futures between 2020 and 2055. These futures represent a wide but plausible range of atmospheric conditions, emissions constraints, and economic, technological, and population growth assumptions. We find that local-level development can vary substantially both by generation type and capacity buildout across these futures. Specific regions of the Western Interconnection are projected to see large amounts of capacity development regardless of the future scenario given local siting drivers. We additionally determine that projected power plant locations are more heavily influenced by the cost to interconnect to the electric grid than the revenue driven by locational energy prices.

Plain Language Summary

The Western United States electric grid is facing major challenges. Electricity demand has become less predictable due to factors like artificial intelligence, large-scale computing, and electrification. At the same time, energy production is shifting toward more diverse sources such as wind and solar. Where new power plants are built in the future will play a critical role in keeping the grid reliable and prepared for future growth.

In our study, we use interconnected models to examine how the types and locations of power plants might change between 2020 and 2055 under different futures. We considered variations in future atmospheric conditions, emissions constraints, regulation, population growth, and other factors. We find that certain regions in the Western US are likely to see substantial power plant development, no matter the future scenario. We also find that certain development costs such as connecting new plants to the grid are a stronger factor in determining new plant locations than the revenue potential of a location. These insights can help planners prepare for growing electricity needs and ensure the grid remains reliable in an evolving landscape.

1. Introduction

The electric grid is facing new reliability challenges due to extreme weather impacts (Doss-Gollin et al., 2023), demand uncertainty due to the onset of artificial intelligence (IEA, 2025a), increased grid interconnection bottlenecks (Gorman et al., 2025), and rising variable renewable energy resource deployment (IEA, 2025b; US EIA, 2025a). Only a decade ago, annual load growth was stagnant, and the US Energy Information Administration (EIA) estimated that it would fall to <1% through 2040 (Sieminski, 2015). This is not the case today. New estimates have projected a rapid upturn in demand whereby total US annual energy use could increase by 53% over the next decade alone (Newell, 2025). Despite the uncertainty in these load growth projections, the expedited deployment of new energy generation resources will be critical to address the growing divergence between projected supply and demand. This is especially true at a time when much of the existing infrastructure is aging (US DOE, 2023) and nearly all recent US power plant retirements consist of dispatchable fossil-fuel resources (US EIA, 2025b) which have historically been used to respond to grid fluctuations. Ideally, long-term energy planning under uncertainty would simultaneously address the opportunities and constraints that influence on-the-ground infrastructure evolution to ensure the feasibility of expansion plans purporting to address complex challenges.

For computational tractability, long-range energy planning models have typically operated at a relatively coarse spatial resolution (e.g., country/region) and therefore have not captured fine-scale constraints and opportunities affecting the infrastructure landscape that could limit the local realization of their plans (Mongird & Rice, 2024). Robust modeling of future grid infrastructure and operations depends on knowing not only the location and magnitude of electricity loads but also how each generator will be interconnected to the larger grid network with an understanding of its physical location and the conditions that will affect its operations (Doss-Gollin et al., 2023; Hadley et al., 2016; Luo et al., 2023). Natural resource availability, evolving local policies, natural hazard risks, energy prices, urban development, and more all present factors that can affect the siting and operational capability of a power plant (Vernon et al., 2023). Multisectoral modeling frameworks that can more accurately represent these dynamic land, human, energy, and water system interactions at decision-relevant scales will be better situated to capture the evolving grid landscape (Doss-Gollin et al., 2023; Mongird et al., 2025).

Most studies of power plant infrastructure potential have focused on methods that use historical data to inform siting decisions through multi-criterion decision making analysis (Khanlari & Alhuyi Nazari, 2022; Kuiper et al., 2022; Rehman et al., 2019; Shao et al., 2020). These methods typically involve using a weighted combination of geospatial factors such as risk (e.g., natural hazards), technological constraints (e.g., irradiance level), socioeconomics (e.g., local attitudes), and environmental feature (e.g., protected areas) in each location to determine siting viability (Khanlari & Alhuyi Nazari, 2022; Shao et al., 2020). These analyses can cover a wide variety of spatially based siting constraints and can capture local-level influences, but they typically utilize fixed inputs in their formulation and do not capture evolving opportunities and restrictions, which can limit their applicability in informing future conditions. Given the increasing uncertainty regarding prospective land use, policy, meteorology, and socioeconomics, using fixed geospatial siting influences may lead to results that are more reflective of historic patterns and may be insufficient for robust long-range energy system planning.

Machine learning (ML) methods are becoming increasingly prevalent in siting decision-making and modeling given their ability to detect complex siting patterns. These methods are often used to determine suitable locations for new development based on existing power plant locations or to identify appropriate multi-criterion decision-making model weights (Sachit et al., 2022; Wang et al., 2021). Additionally, ML methods have been used to evaluate the relative importance of different siting factors and to understand what drives decision-making. Sun et al. (2023), for example, develops an ML-based analysis using existing solar photovoltaic (PV) installations to determine the relative importance of various spatial attributes on chosen siting locations. They find their vegetation parameter and the distance to the power grid to be the dominant spatial predictors of development. These methods, and their ability to be integrated into existing modeling structures, offer ways to gain greater insight from siting results but are also oftentimes restricted to historical development data to train their models.

Some recent research efforts have modeled the future power plant landscape with varying levels of model integration to address dynamic and uncertain land, water, human, and energy system factors. The Nature Conservancy (2023) and Wu et al. (2023), for example, link energy capacity expansion modeling with siting suitability analysis to evaluate the potential for renewable buildouts under alternative siting restriction scenarios. Their research identifies candidate renewable project areas that account for co-located benefits such as agrivoltaics, technoeconomic assumptions, and other criteria to build out future grid portfolios. Similarly, Larson et al. (2021) presents an evaluation of future power plant siting that uses geospatial screening of unsuitable areas to downscale coarse-scale capacity expansion output and choose locations based on least-cost delivered energy under future demand conditions. Both studies integrate capacity expansion models with fine-scale geospatial siting processes and provide examples of methodologies that can be further expanded on by capturing additional dynamic factors that influence developer decision-making. Despite these advancements in energy infrastructure modeling, previous research has not explored how the future landscape of renewable and non-renewable generation in the US would evolve differently under alternative future atmospheric conditions, emissions constraints, technology change, and population and economic trends and how fine-scale geospatial considerations would affect that evolution. Our research addresses these gaps.

We have developed an open-source integrated modeling framework that simulates high-resolution energy-water-land-urban dynamics, vulnerabilities, and adaptation within the US under acute and long-term human-Earth system stressors. This research effort lies within the relatively recently coined field of multisector dynamics (Reed et al., 2022), but our models and model coupling strategies have been in continuous development and application for more than ten years, providing insights into competing land uses, water scarcity, electricity system grid stress, and urban heat stress (see, e.g., Kraucunas et al. (2015); Hejazi et al. (2015); Voisin et al. (2016); Chen et al. (2019); Zoraghein and O'Neill (2020); Karimi et al. (2022); Vahmani et al. (2022); Vernon et al. (2023); Ssembatya et al. (2024); Gold et al. (2024); Akdemir, Mongird, Bracken, et al. (2025)). For this paper, we integrate selected models from the larger framework that dynamically project the evolution of the power plant infrastructure landscape within the US Western Interconnection 2020-2055 across a wide yet plausible range of scenarios. We use a factorial scenario design to explore the impacts of uncertainty in future emissions constraints and resulting atmospheric conditions (including global circulation model [GCM] temperature sensitivity) as well as uncertainty in population

and economic growth. The Western Interconnection includes eleven US states (Arizona, California, Colorado, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming). We address the following science questions in this research: (1) Which factors in our scenario design have the largest impact on the evolution of the power plant landscape in the US Western Interconnection?; (2) Which subregions see consistent power plant infrastructure development regardless of future pathway?; and (3) Which parameters in the siting model are most influential, and how does the level of influence vary by generating technology and future scenario?

The remainder of this paper is structured as follows. We begin with our methods, including an overview of the integrated modeling framework, details on the power plant siting model, and our spatial analysis methods. Next, we present our results, beginning with the projected capacity expansion by Western US state and the associated power plant siting results. This is followed by an analysis that identifies regions of the US Western Interconnection that see both high levels of power plant development and consistent development across our scenarios. Last, we discuss the results of an ML analysis that quantifies the relative influence of power plant siting model factors on infrastructure location choice. We finish with a discussion of limitations, future research extensions, and concluding remarks.

2. Methods

This research evaluating the potential future power plant landscape is part of a larger set of experiments exploring the evolution of electricity system grid stress within the US under varying influences. Grid stress is defined as a period when the costs of meeting electricity demand increase substantially, or outages occur because demand cannot be met. Grid stress can occur due to demand-side, supply-side, or interacting supply and demand side factors. For example, periods of high or rapidly increasing electricity demands (i.e., loads) combined with insufficient electricity supply can lead to grid stress. Since electricity demand needs to be met instantaneously at the time and place it occurs, understanding when, where, and what new supplies (e.g., new power plants) will be built is critical to modeling future grid stress. The following subsections provide an overview of our integrated modeling approach to explore grid stress, a description of the power plant siting model, and a summary of our spatial analysis methods.

2.1. Integrated Modeling Approach

Our integrated modeling framework for grid stress accounts for human-Earth system dynamics affecting the electricity system and has been previously published in Akdemir, Mongird, Bracken, et al. (2025). The general approach is to simulate the impacts of wide-ranging scenarios of future weather (jointly with emissions constraints) and socioeconomic conditions on energy-water-land interactions using a global change analysis model, and then to use those simulation results for the energy sector as boundary conditions for finer scale, process-based modeling of US electricity supply (including power plant locations), demand, and operations for each scenario.

For the future weather scenarios, we performed perturbed thermodynamics experiments (Jones et al., 2023) to reproduce historical CONUS meteorology under a wide range of future thermodynamic forcings derived from the

Coupled Model Intercomparison Project 6 (CMIP6) (Eyring et al., 2016; O'Neill et al., 2016). We provide a high-level summary of the Jones et al. (2023) approach here. First, we divided selected GCMs into hotter versus cooler groups based on an analysis of their temperature sensitivity over CONUS. Then we derived monthly temperature and humidity signals over the CONUS from these two groups' simulations achieving a radiative forcing of 4.5 W/m² and 8.5 W/m² by 2100, representing moderate emissions constraints and moderately hotter and drier conditions versus no emissions constraints and severely hotter and drier conditions, respectively. Using the ERA5 30-km historical reanalysis (1980-2019) (Hersbach et al., 2020) over the CONUS as our historical baseline, we next performed dynamical downscaling to 12-km with the Weather Research and Forecasting model (WRF; Skamarock et al. (2019)), applying the time-evolving monthly thermodynamic signals to the ERA5 30-km boundary conditions. In all, we created four future (2020-2059) scenarios of hourly meteorology, each with a hotter and cooler variant.

For our socioeconomic drivers, we used population and economic growth projections consistent with the Shared Socioeconomic Pathways (SSP) 3 and 5 (O'Neill et al., 2020; Zoraghein & O'Neill, 2020). These socioeconomic scenarios provide strong contrasts within the US: lower population and economic growth under SSP3 versus higher population and economic growth under SSP5. Following the ScenarioMIP structure (O'Neill et al., 2016) and related research (Ren et al., 2018) demonstrating that emissions consistent with an 8.5 W/m² radiative forcing can be achieved under SSP3 with a small increase in GDP growth rates, we pair each of our four future weather scenarios with SSP3 and SSP5 to produce an eight-scenario factorial design (Table 1).

Table 1. Scenarios of emissions constraints, global circulation model temperature sensitivity, and population and economic growth over the CONUS.

Scenario Name*	Emissions Constraints	Degree of Hotter and Drier Weather	GCM Sensitivity	Population and Economic Growth
rcp45cooler_ssp3	Moderate	Moderately	Cooler	Low
rcp45cooler_ssp5	Moderate	Moderately	Cooler	High
rcp45hotter_ssp3	Moderate	Moderately	Hotter	Low
rcp45hotter_ssp5	Moderate	Moderately	Hotter	High
rcp85cooler_ssp3	None	Severely	Cooler	Low
rcp85cooler_ssp5	None	Severely	Cooler	High
rcp85hotter_ssp3	None	Severely	Hotter	Low
rcp85hotter_ssp5	None	Severely	Hotter	High

* We use the acronym "rcp" (Representative Concentration Pathway) for brevity in our scenario names—this does not refer to the use of CMIP5 simulations. As described in the text, our perturbed thermodynamics simulations are derived from CMIP6 simulations.

Figure 1 shows how the exogenous scenario forcings and each model's simulation results propagate across the framework. We use scenario-forced simulations from the Global Change Analysis Model with subnational detail for the US (GCAM-USA) (Binsted et al., 2022; Calvin et al., 2019) to generate the energy system expansion at the state level and to provide inputs for multiple downstream models. The Helios model (Zhao et al., 2024) provides weather impacts on energy demands and capacity needs for GCAM-USA by calculating population-weighted heating and cooling degree-hours (HDH and CDH) for each US state using the hourly gridded weather data and population projections. State-level HDH and CDH are aggregated into 24 monthly day/night load segments (12

months $\times$ 2 time-of-day periods) plus one super-peak segment representing the 10 highest electricity load hours of the year to create load duration curves for use within GCAM-USA. For the generation capacity expansion, at each five-year GCAM-USA timestep, we use the highest super-peak from all previous years (plus a reserve margin) to determine capacity needs. This is consistent with standard industry practice for capacity expansion modeling (Bystrom, 2022). For GCAM-USA's simulation of annual energy generation at each timestep year we use the load duration curve from the year within the time step that best reflects average annual energy use within each 5-year period. Note that all GCAM-USA simulations using the moderately hotter and drier projections include emissions constraints consistent with achieving a radiative forcing level of 4.5 W/m^2 by 2100 (Calvin et al., 2017). GCAM-USA's simulations forced by the severely hotter and drier scenarios do not include any emissions constraints.

The SSP assumptions are implemented in GCAM-USA as follows. SSP3 and SSP5 differ in population, GDP, technology costs, and fossil fuel availability and preference, which shape electricity system outcomes. SSP5's higher domestic population and GDP growth in the US drive greater electricity demand across buildings, transportation, and industry. However, SSP5 also assumes more abundant fossil fuel extraction and less expensive carbon capture and sequestration (CCS), making gas and coal with CCS cost-competitive regardless of the emissions constraint level. With emissions constraints, both SSPs are pushed toward lower-carbon technologies, but SSP5's less expensive CCS means more fossil fuel generation with CCS compared to SSP3. Without emissions constraints, SSP5's less expensive fossil fuels have even more room to dominate, while SSP3's relatively more expensive CCS continues to favor renewable deployment. As a result, differences in electricity system composition across our scenarios reflect the interaction between levels of emission constraints and which technologies are economically competitive under each SSP. More detailed information on SSPs in GCAM-USA can be found in the GCAM-USA documentation (JGCRI, 2026).

The Total Electricity Load (TELL) model (Burleyson et al., 2025; McGrath et al., 2022) models hourly electricity loads as a function of weather, gridded population, and GCAM-USA energy demand projections. The reV model (Maclaurin et al., 2021) produces hourly solar and wind resource generation profiles based on the weather scenario. These models' outputs feed into the high-resolution modeling of transmission line capacity additions and electric grid operations, modeled by the Transmission Expansion Planning (TEP) (Akdemir, Mongird, Kern, et al., 2025) and Grid Operations (GO) (Akdemir et al., 2024) models, respectively. Wholesale electricity prices produced by the GO model inform the modeling of future electricity system infrastructure locations carried out by the Capacity Expansion Regional Feasibility (CERF) (Vernon et al., 2021). More details on the model interactions shown in Figure 1 appear in Akdemir, Mongird, Bracken, et al. (2025), a study of future electricity system grid stress across our scenarios. In this paper, we focus on the CERF power plant siting results.

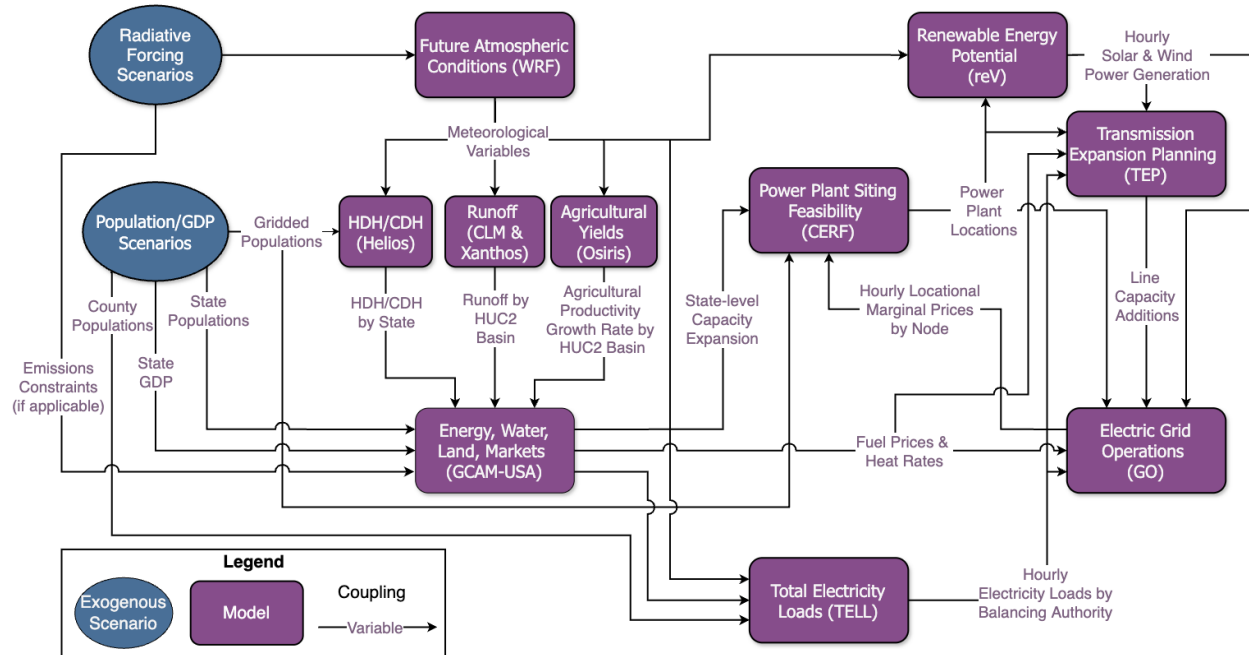


Figure 1 Diagram of the integrated modeling approach. Blue ovals represent exogenous scenarios, purple rectangles represent models, and connecting arrows represent model couplings.

2.2. The Capacity Expansion Regional Feasibility (CERF) Power Plant Siting Model

CERF is an open-source geospatial power plant siting model that takes a regional capacity expansion output, typically at a state-level (e.g., from GCAM-USA), and determines site-specific power plant locations on a 1 km² grid based on geospatial constraints and economic drivers (Mongird et al., 2025; Vernon et al., 2021). The model does not claim to provide an optimized regional siting plan but rather to identify feasible siting plans constrained by land suitability and informed by locational economics. The model sites renewable and non-renewable utility-scale generation technologies as well as varying technological configurations (e.g., turbine type, cooling type, with/without CCS)—currently 85 unique technology configuration types. CERF contains hundreds of high-resolution geospatial siting suitability layers representing siting exclusions specific to each technology, scenario, and year combination (Vernon et al., 2023). These include areas such as protected lands, population density, cooling water availability, slope exceedance, renewable resource availability, and proximity to infrastructure. Geospatial constraints used for population density are consistent with the SSP scenarios used in the broader experiment and evolve over time. Vernon et al. (2023) provides a validation of how CERF's geospatial layers perform against historic power plant siting locations.

CERF's economic component is an algorithm addressing the costs and benefits that influence siting decisions. At each timestep and for each suitable grid cell/technology combination, CERF calculates an annualized net locational cost based on the tradeoff between the technology-specific cost to interconnect to the electric grid from that location and the technology-specific net operational value (i.e., the difference between the power plant's variable operating costs and projected revenues based on locational marginal prices [LMPs]) in that location.

Transmission interconnection cost is determined as the cost of building new transmission to the nearest substation. Gas pipeline interconnection cost (for natural gas-fired plants) is determined as the cost to build a new gas pipeline to the nearest gas pipeline network. Both costs are calculated based on straight line (Euclidean) distances. Total interconnection cost is the sum of these two components. CERF does not consider plant capital costs in its economic algorithm because it is siting power plants that have already been “bought” by each US state simulated in GCAM-USA as part of its capacity expansion optimization. In other words, CERF is not re-simulating the expansion plan but rather determining feasible locations within each state for the power needs pre-determined by GCAM-USA. Vernon et al. (2021) and the associated model documentation describe the economic algorithm in greater detail. The LMPs used by CERF are derived from a grid operations model (e.g., GO) which determines them based on the system’s marginal production cost, which reflects fuel prices, the resource mix, unit dispatchability, and transmission congestion. Within CERF, the LMP used in the project revenue calculation for each technology depends on its annual average capacity factor (i.e., the fraction of hours in operation during a year) and dispatchability. For example, a gas turbine with a 30% capacity factor is utilized during peak hours of the year when LMPs are typically higher. In this case, we use the average of the top 30% of LMPs across the 8760 hours of the year for each grid node to determine the potential operating value at each location. A solar farm, on the other hand, cannot be dispatched as needed and only operates during daylight hours. Therefore, the operating value of a solar farm in CERF is dependent on the LMPs during daylight hours at each location. The annual LMPs vary by timestep and scenario as the resource mix changes, as weather affects renewable output, and as demand evolves under changing weather and socioeconomic conditions. The LMPs used in CERF at each timestep are obtained from a GO simulation that uses the projected demand for the timestep along with the previous timestep’s generation resources. This process is intended to represent the marginal value of new generation in different locations and to be consistent with on-the-ground regional-scale infrastructure planning processes that typically take projections of demand growth into account. These look-ahead LMPs provide a stronger signal for where new generation would be valuable compared to using LMPs from the previous timestep.

As CERF processes a given expansion plan, it sites technologies to minimize net locational costs. In many cases, a grid cell may be suitable for multiple technology types. In this case, the technologies will compete for the location based on their net locational costs. The CERF siting output, in the form of technology-specific capacity in each grid cell, individual siting coordinates, and grid interconnection locations, is passed to both the renewable generation model (reV), to create hourly solar and wind profiles based on solar and wind locations and corresponding weather data, and to the grid operations model (GO). See Campbell et al. (2024) and Bracken et al. (2025) for a thorough discussion of the wind and solar generation modeling. GO updates its generation resources for the timestep with the CERF sitings and reV results and runs again to produce the final operational results for the timestep (i.e., GO is run twice at each time step: once in a preliminary mode to provide the look-ahead LMPs to guide CERF sitings and again once the CERF siting has been completed).

In addition to siting new capacity, CERF also accounts for the impact of power plant retirements on a grid cell basis. In our framework, GCAM-USA provides the remaining aggregate capacity available by technology type and state compared to the base year infrastructure (i.e., 2015). GCAM-USA decreases fossil fuel-fired capacity over

time following retirement projections from existing data sources and through an economic retirement schedule that captures fuel prices and operating life of the technology. To translate this into site-specific retirements, we use 2015 power plant data (US EIA, 2016) consistent with GCAM-USA's base year assumptions and the incremental state and technology-level decrease in capacity from GCAM-USA to systematically retire pre-existing facilities from oldest to newest. A facility is only retired once the amount of reduced capacity for a given state and technology combination from GCAM-USA surpasses the capacity size of the given facility. Solar, wind, and geothermal facilities follow an alternative schedule and retire at the end of their assumed operating life after 2015. Once a power plant retires, its location becomes available for siting new infrastructure within the CERF model. Hydroelectric generation is assumed to remain static across the experiment time horizon with no new development and no facility retirements.

The version of GCAM-USA used in our integrated experiment does not model energy storage directly. However, we do include energy storage in our overall experiment by assuming each grid interconnection point that receives new solar or wind development is also paired with battery energy storage capacity equivalent to half the interconnected renewable capacity. The 50% capacity assumption was determined by evaluating existing energy storage and renewable projects (LBNL, 2024). No additional siting considerations are considered for energy storage as it is not a technology included in the power plant siting model. For more details on the storage modeling within this experiment, please refer to Akdemir, Mongird, Bracken, et al. (2025).

2.3. Spatial Analyses

2.3.1. Scenario and Siting Intensity by Region

In addition to evaluating siting trends at the 1 km resolution, we also analyze spatial distribution and siting patterns by technology type across scenarios to identify regional and locational trends. Specifically, we take individual power plant coordinates across all scenarios and timesteps and map them into a 50 km resolution grid (i.e., each grid cell representing a 50 km $\times$ 50 km area). This resolution was chosen for this analysis because it is coarse enough to aggregate individual power plant sitings into meaningful regional clusters while still being fine enough to differentiate sub-regional patterns across the Western Interconnection. From there, we calculate two key metrics for this analysis per grid area: the average amount of capacity installed across all scenarios and the number of scenarios in which the grid cell experiences sitings. This is completed for various groupings of technologies to determine trends by generation type including non-renewables, solar PV, wind, and all technologies in aggregate. From this analysis, we identify regions that (1) have no sitings at all, (2) see a large amount of capacity on average but only within a small number of scenarios, (3) see a small amount of capacity on average but in every scenario, or (4) see both high average capacity installation and sitings in every scenario. This analysis, conducted from our siting model output, allows us to spatially identify deployment hotspots from a land use and societal perspective and highlight where future energy infrastructure may concentrate and where associated implications for land and human systems are most relevant.

2.3.2. Feature Importance of Key Siting Parameters

It is not only the regional differences in power plant siting results that are of interest but also the importance of various factors that the CERF model uses to determine the ultimate siting location. Two primary economic factors determine the net locational cost in each location: the cost of grid interconnection and the LMPs. For the former, this includes the \$/km cost to build new transmission from the siting location to the nearest substation based on straight-line (Euclidean) distance. Natural gas-fired facilities have an additional component to their interconnection cost (IC) in the form of a new gas pipeline to connect to the existing gas pipeline network. As described earlier, LMPs are a key input to the net operating value; they are dynamic and depend on the output of the grid operations model and the generation profile of the technology (e.g., peaking vs. baseload, daylight hours for solar PV, and dispatchability). The ultimate net locational cost is determined by the tradeoff between these two factors in each location. We investigate the relative influence of IC and LMP on whether a particular grid cell is selected for siting using a random forest (RF) classifier (Breiman, 2001). Additionally, to interpret the model's feature contribution, we employ SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017).

To run the analysis, we built an extensive dataset that contains identifying features for every 1 km² grid cell in the analysis space, including corresponding IC, LMP, and siting suitability for each technology, timestep, and scenario combination in our experiment. The LMPs correspond to data generated from the GO model to site new power plants for the given timestep. These datasets are then filtered to remove grid cells that are unsuitable for siting for the given technology in the given timestep/scenario combination. These areas are removed to ensure we are analyzing choices between viable grid cells only. Unsuitable grid cells never receive sitings and would heavily skew feature importance towards not siting if they were retained.

Since there are many more grid cells in the dataset that are suitable but did not receive sitings compared to suitable grid cells that did see sitings, we rebalance our dataset by randomly sampling non-sited grid cells for every sited grid cell by technology type, timestep, scenario, and state combination (Figure 2). Rebalancing is necessary to avoid disproportionately skewing the datasets with areas that did not have positive siting values, which would bias the classification towards the majority (Kotsiantis et al., 2005). We tested both a 1:1 resampling ratio and a 2:1 resampling and found the latter performed with higher accuracy on average. Based on the compiled datasets, only technologies with many samples (i.e., several hundred) are retained for the analysis. For example, utility scale biomass generation sees very few sitings across our experiment time horizon and would have a total sample size of $n < 50$ in some cases, so it is not included. The final list of technologies in the analysis is natural gas combined cycle with recirculating cooling, natural gas combustion turbine, solar PV, and onshore wind at 80m, 100m, 120m, and 140m hub heights, individually. In total, we build 48 individual models (six technologies $\times$ eight scenarios) to analyze feature influence.

Example 1 km² resolution siting region for given technology

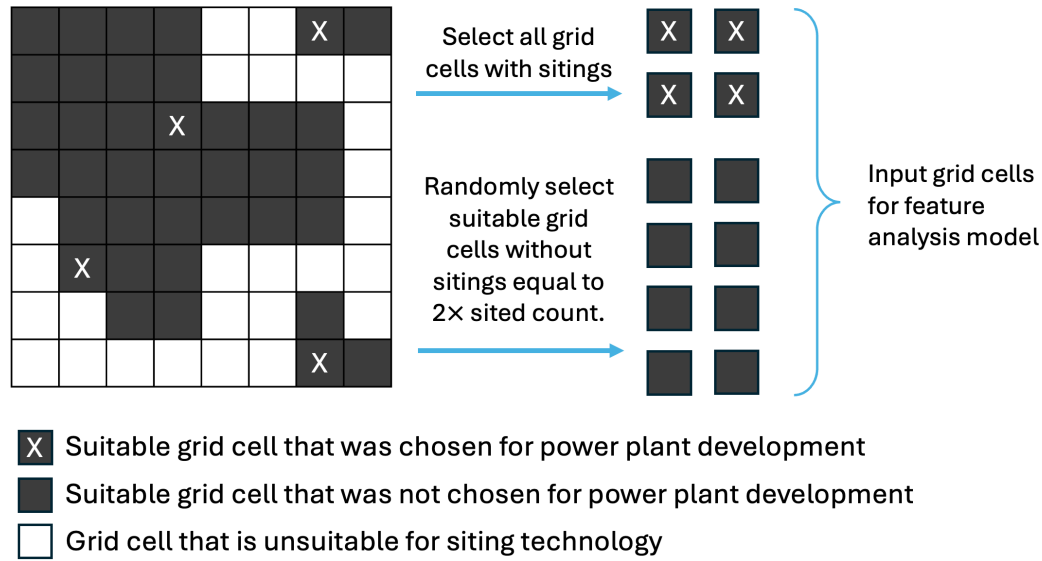


Figure 2. Feature analysis data sampling method diagram. Each individual grid cell represents a 1 km x 1 km area.

Using our balanced datasets, we run a RF classifier model using IC and LMP as features for each technology and scenario combination using an 80/20 training and testing split, a common choice for balancing model training and evaluation. Gini impurity is chosen over entropy for determining tree splits for computational efficiency. Hyperparameters (e.g., number of estimators, max tree depth, etc.) are tuned for each model using cross-validation. The best performing configuration is chosen for each model based on accuracy (i.e., the proportion of correct predictions). The number of folds used in the cross-validation, which represents the number of equally sized subsets a training dataset is split into to create train/test data subsamples, depends on the dataset size. For smaller datasets ($n < 1000$), ten folds are used to boost the size of the training set. For all other datasets, five folds are used for faster computation. The RF classifier constructs decision trees using a random selection of dataset rows (i.e., bootstrapping). These trees independently predict the ultimate class (sited or not) given the values of its input features (IC and LMP). The final prediction of whether an individual grid cell receives a siting is based on a majority vote across the individual trees. Overall, the models reported an accuracy range between 80-99% with >75% of models performing with 94% accuracy or higher. Classification reports including precision, recall, and f1-scores for all 48 models are provided in the output dataset associated with this manuscript (Mongird, 2025).

To interpret the relative influence of each feature on power plant siting, we pair our RF models with SHAP. SHAP determines the degree to which a feature moves a prediction above or below the average prediction for the background dataset. SHAP offers advantages for interpretability both on a per grid-cell prediction basis as well as informing us of the magnitude of the influence of each feature. We may suspect, for example, that higher IC values decrease the likelihood of a location being chosen for siting, but SHAP can show us whether that is true and to what degree it is true. We use the mean of the absolute value of individual SHAP values to infer the comparative influence of IC vs. LMP for each technology on siting outcome and to determine the degree to which different scenarios affect that influence.

3. Results & Discussion

Our analysis focuses on projected power plant sitings (2020-2055, 5-year timesteps) at the local level in the Western Interconnection across our scenarios (Table 1). We begin with a state-level overview of the capacity expansion output from GCAM-USA and then present maps of the CERF-generated power plant landscape in the Western Interconnection as of the year 2055. This is followed by an analysis of local siting patterns where we identify subregions that see consistent and/or high levels of power plant development across scenarios. Finally, we discuss the results of our RF classifier + SHAP feature analysis, which explores the importance of IC vs. LMP in siting decisions by technology and scenario.

We find that the uncertainty in the future weather (which incorporates the presence of emissions controls or not) dominates the uncertainty in SSP with respect to impacts on power plant siting. As a result, we present results here for the moderately versus severely hotter and drier weather futures under SSP5 only and focus on the impacts of the weather/emissions scenarios along with the impacts of the hotter and cooler variants. Figures for all SSP3 scenarios are provided in the Supplemental Information (S1, S3-5).

3.1. State-level Capacity Expansion Results under Alternative Futures

As shown in Figure 3, the future generation resource mix within and across the eleven states in the Western Interconnection varies considerably by scenario and over time.

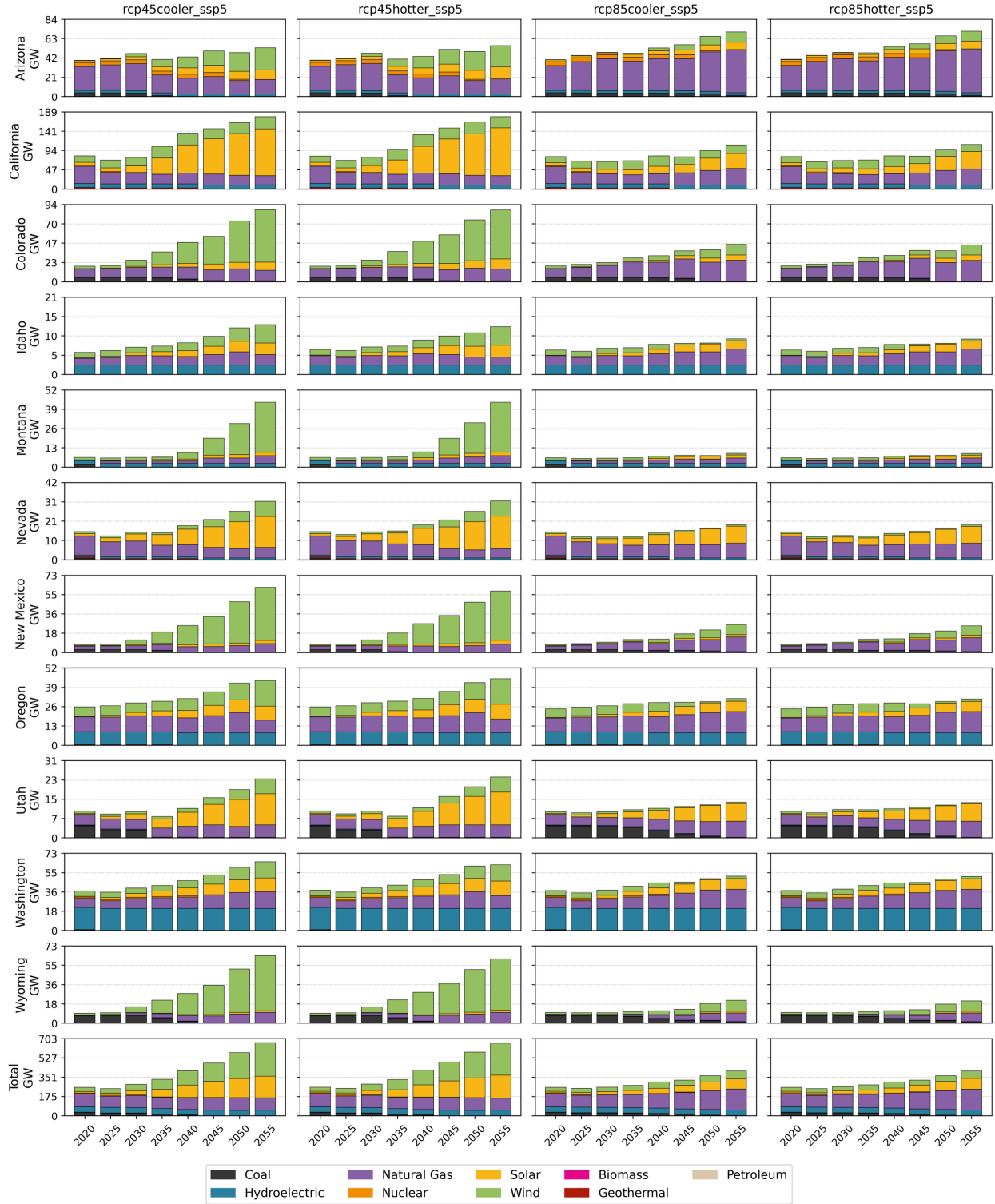


Figure 3 Total capacity in gigawatts by state and timestep across the four SSP5 experiment scenarios. The first eleven rows represent state capacity expansion while the last row represents the total capacity expansion across the entire US Western Interconnection over time. Scenarios include the combination of a moderately hotter and drier future on average across the US with moderate emissions constraints (rcp45), and a severely hotter and drier future on average across the US with no emissions

constraints (rcp85), respectively; and two GCM sensitivity conditions, cooler represents a relatively cooler prediction, whereas hotter represents a relatively hotter prediction.

We see from Figure 3 that, despite the relatively cooler temperatures in the moderately hotter and drier scenarios, every state except Arizona has larger capacity deployments in these scenarios compared to the severely hotter and drier scenarios. The majority result is due to the higher degree of economy-wide electrification and renewable development needed to meet the emissions constraints, with the subsequent need for more capacity. In Arizona, the highest overall demands and capacity deployments occur in the severely hotter and drier scenarios. This is primarily due to the significant base year penetration of air conditioning responding to dramatic temperature increases and a higher overall demand in the buildings sector.

Figure 3 also demonstrates state-level diversity in the expansion. For example, under the moderately hotter and drier emissions-constrained scenarios, California, Nevada, and Utah's capacity expansions are primarily solar power, while Colorado, Montana, New Mexico, and Wyoming's expansions are primarily in wind power. On the other hand, without emissions constraints in the severely hotter and drier scenarios, capacity deployment shifts primarily to natural gas-fired generation. GCM sensitivity has a lower impact on the capacity expansion across the Western Interconnection, though small differences appear at the state level in years closer to 2055.

We can also identify differences in retirement schedule of pre-existing fossil-fired generation in Figure 3. Under our rcp45 scenarios, phase-out of pre-existing coal-fired generation occurs over a decade sooner in some states. Pre-existing natural gas-fired generation follows a similar pattern. To compare the capacity expansion in Figure 3 to forward-looking estimates of future grid generation by the Western Interconnection, please refer to Figure S2 in the supplemental information which provides estimated capacity by state in 2034 (WECC, 2025).

3.2. Local-scale Power Plant Landscape Evolution Across Alternative Futures

The CERF power plant siting model (see Methods) takes the GCAM-USA state-resolution capacity expansion plans (Figure 3) and translates them to individual power plant sitings at a 1 km² resolution for each of our scenarios (Figure 4). Figure 4 shows on the ground power plants by 2055, mostly consisting of new development since most of the pre-existing generation retires prior to 2055. These results demonstrate the differences in where projected capacity expansion is expected to be realized given siting constraints and grid conditions under each scenario. Figure 4 shows several diverging clusters of technology deployments depending on the weather/emissions and GCM sensitivity combination.

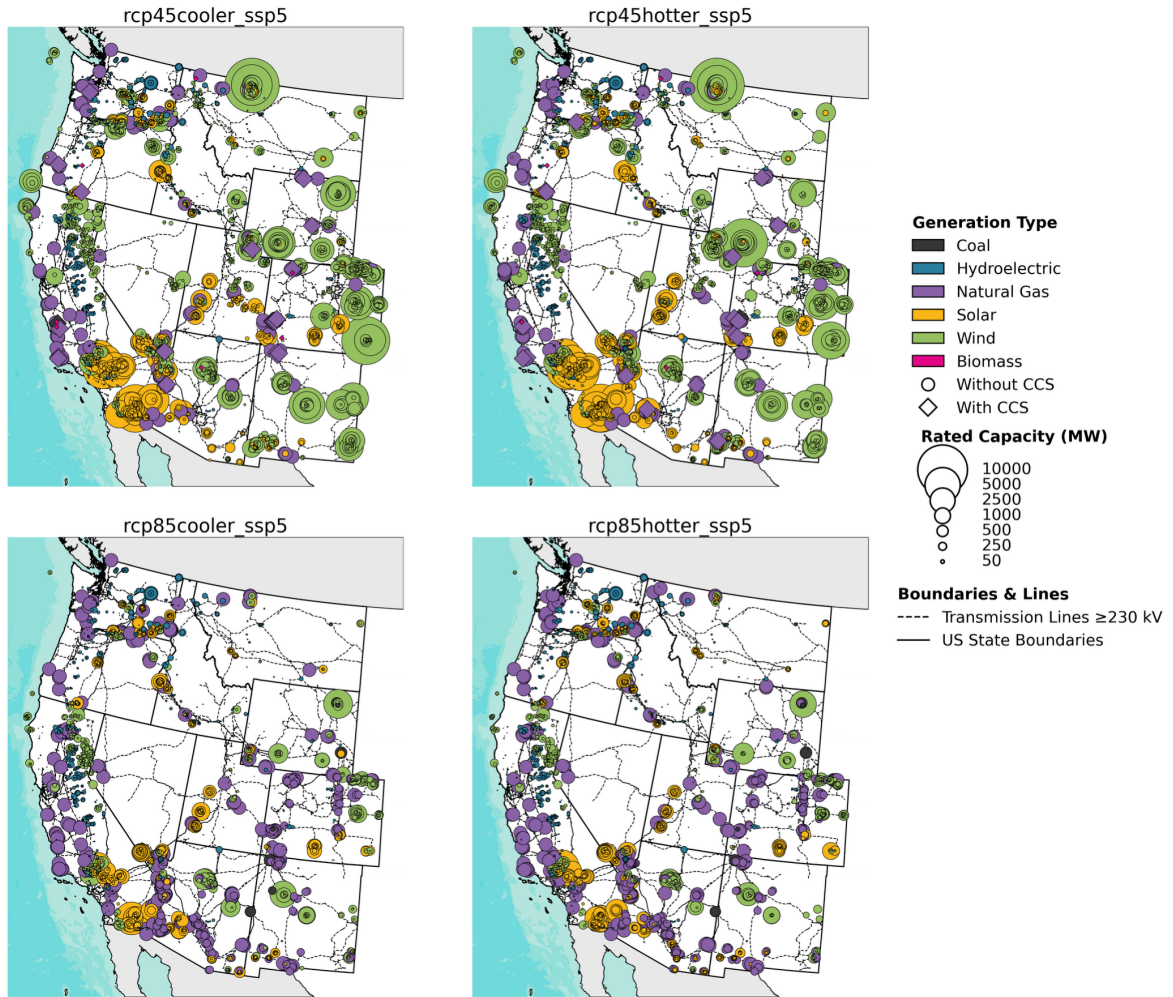


Figure 4 US Western Interconnection power plants locations 2055 under four projected futures. Markers are colored by generating technology type and sized by rated capacity. A diamond marker, as opposed to a circle, indicates that the power plant contains carbon-capture sequestration.

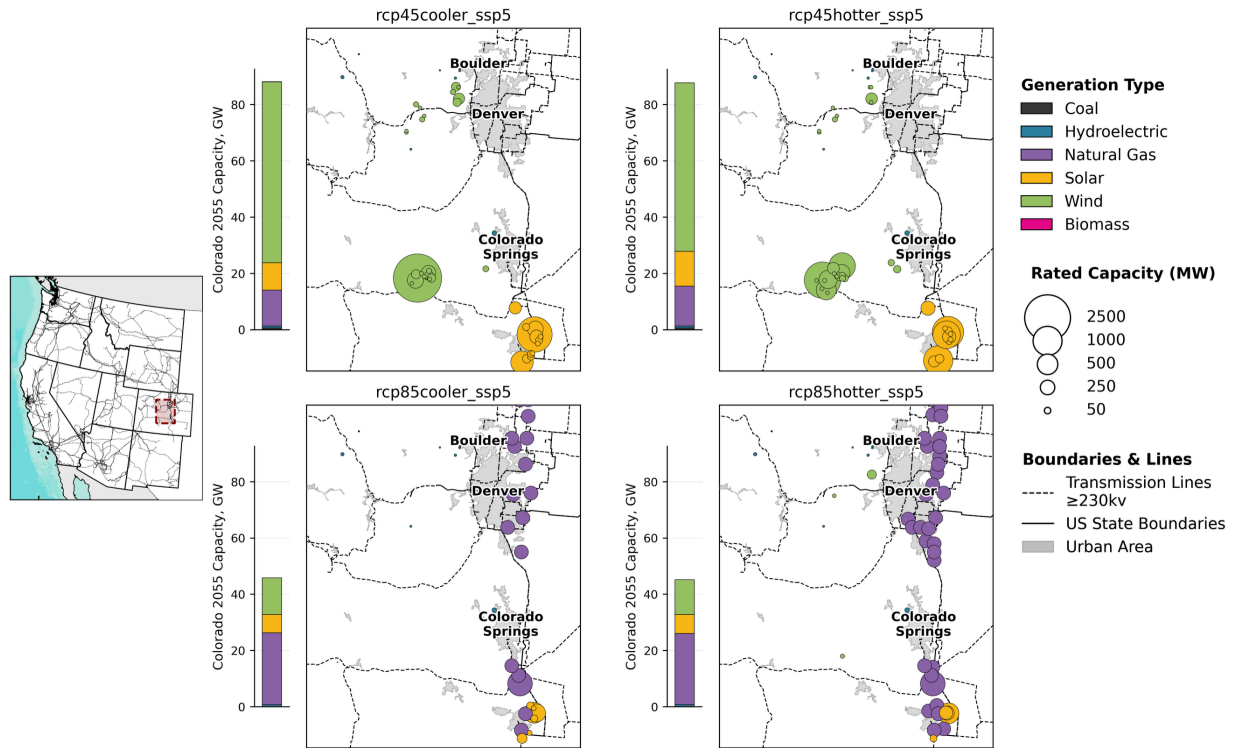
Our results indicate that our weather/emissions scenario uncertainty is the primary driver towards determining the spatial evolution of the power plant landscape. The greater share of renewables under our rcp45 scenarios compared to the larger share of gas buildout under the rcp85 scenarios dramatically impacts the spatial distribution of energy infrastructure across the Western Interconnection. The scale of renewable deployment under emission-constrained futures pushes large amounts of wind siting into regions that have not historically seen high levels of development, such as Montana and northern California. In comparison, scenarios without emissions constraints have lower total capacity by 2055, with a greater share composed of dispatchable and higher capacity density generation. This produces large differences in land use for power plant infrastructure over time. Ultimately, we find the concentration of power plants in specific regions is strongly driven by the different policy and technological assumptions across our scenarios. This reveals that regional and local land use impacts from power plant development are not uniformly distributed across energy futures.

Figure 4 also demonstrates how siting patterns can emerge based on generation technology type. Wind development is dispersed across states under each scenario but appears in large clusters in northern California, Colorado, Arizona, and Wyoming. The largest amount of wind development by individual farm capacity occurs in northern Montana under the rcp45 scenarios when there is more than triple the wind capacity deployed in the state compared to the rcp85 scenarios. Under rcp85, wind farms are projected to be comparatively smaller in terms of farm capacity, and the largest capacity clusters appear more equally in the states of Colorado, Wyoming, and New Mexico. Figure 4 shows that solar PV deployment principally occurs in the southern California region but there are additional clusters within Nevada, Arizona, and Utah across all scenarios. The power plant siting model sites facilities in areas capable of meeting the capacity factor set by the capacity expansion model in each given simulation year. Most sitings for solar PV occur in areas near where historical development has taken place given the high irradiance levels in the southwest region and adequate suitable land.

These siting results can be compared to recent literature. In Wu et al. (2023), the deployment of onshore wind across the Western US appears consistent with many of the locations in this analysis, however, notable differences appear in specific regions. In New Mexico, for example, our wind sitings are predominantly located in the western part of the state for the rcp85 scenarios, whereas the eastern region of the state only sees larger development under the rcp45 scenarios. In Wu et al. (2023), wind sitings are more consistent with the latter across scenarios evaluated. Our results can also be compared alongside studies that model decarbonization scenarios such as Larson et al. (2021). While both studies identify consistent solar and wind development regions, we find that the size of wind and solar developments in each location differ by scenario. Additionally, by capturing scenarios without emissions constraints, we can evaluate futures that have a larger buildout of gas-fired generation and evaluate outcomes that decarbonization-only analyses may miss. Only by evaluating over a range of futures can we determine which areas are sensitive to alternative socioeconomic, policy, and atmospheric conditions. Studies that use fixed climate or socioeconomic assumptions may not be as informative for long term energy planning and projection of local level land use impacts.

CERF's 1 km² resolution allows us to explore the potential local impacts of siting. For example, Figure 5 shows the projected power plant landscape in 2055 in the Denver, Colorado area under each of the four scenarios. This local-scale map reveals the community-level differences in plant infrastructure that are less apparent from a higher perspective. Note the disparities in both the technology mix and the amount of capacity deployed in each scenario. Both rcp45 scenarios show large solar and wind facilities outside the major urban areas, whereas the rcp85 scenarios demonstrate a sizeable amount of natural gas deployment within and around the urban areas and little to no solar or wind. While the rcp45 scenarios have larger implications for land conversion in more rural areas, the rcp85 scenarios demonstrate how land use and emissions from utility-scale fossil-fuel generation resources in urban areas may increase compared to what is installed today. Figure 5 also demonstrates how a hotter GCM sensitivity has a small impact in the Denver area under rcp45 but results in a substantial increase in gas deployment around urban areas under rcp85, despite having nearly equivalent capacity expansion plans for the state. This is due in part to wholesale electricity price differences across the region under cooler vs. hotter GCM sensitivity that incentive siting towards different regions.

459



460

461 *Figure 5 Detailed power plant projections in 2055 in the Denver, Colorado area. Markers are colored by generating technology*
 462 *type and sized by rated capacity.*

463 3.3. Siting Density Variability Across Scenarios

464 By further evaluating the individual power plant siting locations across scenarios, we can identify subregions
 465 that see (1) no new development under any pathway, (2) consistent deployment across different scenarios, (3) large
 466 amounts of capacity deployment, or (4) both consistent and large deployment. By determining regions that are
 467 projected to always see deployment of new capacity, as well as areas that rarely or never see new capacity, this
 468 analysis may provide useful insights for regional planning of new infrastructure.

469 Figure 6 shows these siting intensity results using a bivariate choropleth plot where individual 1 km² siting
 470 locations for new infrastructure are mapped to a 50 km resolution grid. This resolution is coarse enough to aggregate
 471 individual power plant sitings into meaningful regional clusters while still being fine enough to differentiate sub-
 472 regional patterns across the Western Interconnection. The bivariate coloring indicates both the number of scenarios
 473 (shades of blue) in which the particular grid cell sees new power plant development of any size and the average
 474 amount of capacity (shades of red) across all four scenarios that is deployed in that grid cell. Bright red grid cells,
 475 for example, see >5000 megawatts (MW) of new capacity deployed on average, but only in one out of four
 476 scenarios. Bright blue grid cells, on the other hand, indicate that new power plants are deployed under all scenarios,
 477 but in lower capacity quantities. Dark purple grid cells indicate areas that see higher levels of capacity deployed on

average as well as deployments in every scenario. This analysis was run for all technology types in combination as well as for non-renewables, onshore/offshore wind, and solar PV individually.

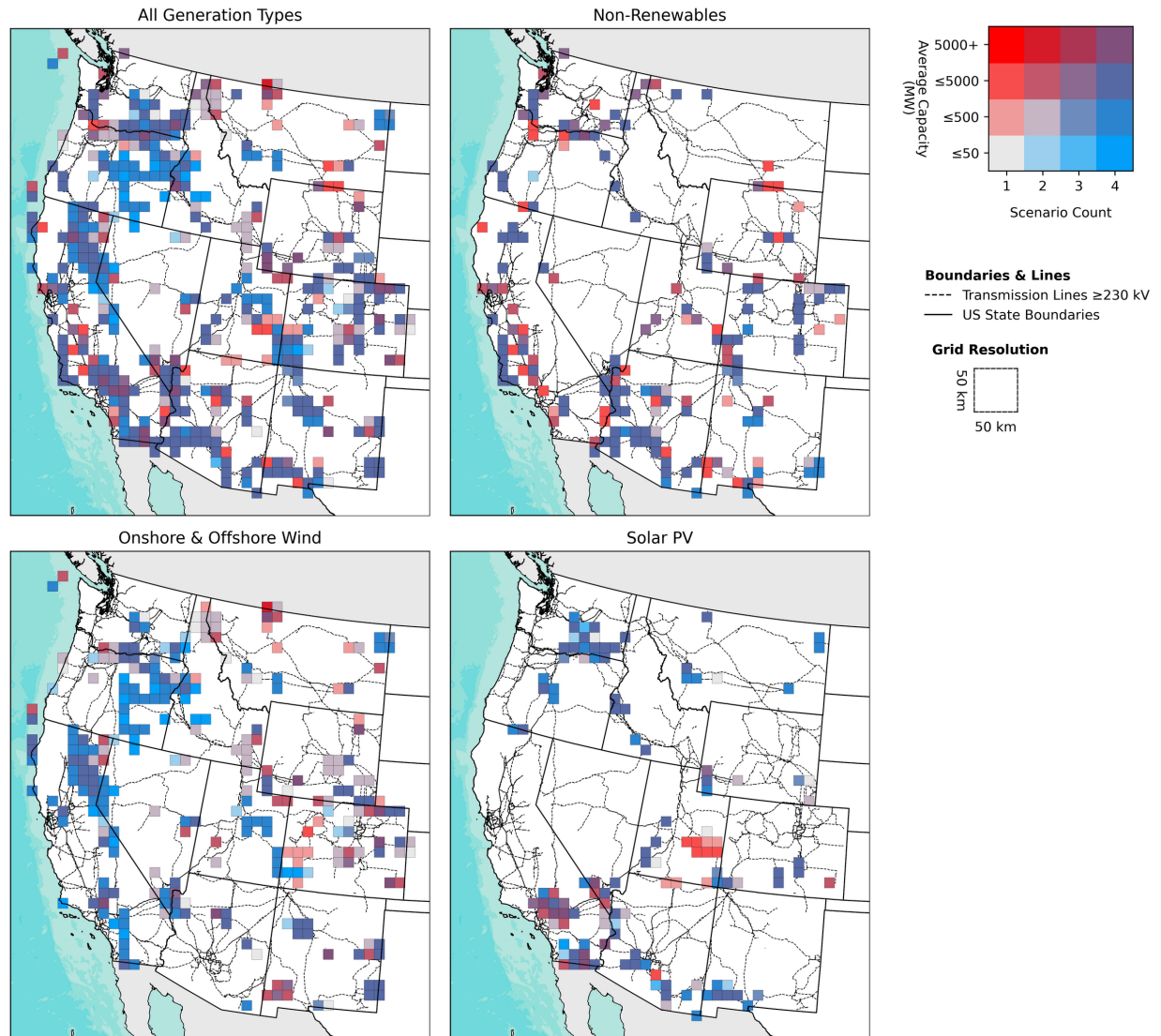


Figure 6 Bivariate choropleth of scenario count and average capacity deployed on a 50 km resolution grid for all generation types (upper left panel), non-renewables (upper right panel), onshore/offshore wind (lower left panel), and solar PV (lower right panel). Scenario count indicates the number of scenarios (out of the four scenarios with SSP5) that the 50 km resolution grid cell saw capacity deployment. The average capacity corresponds to the average MW of new capacity deployed in the grid cell across all scenarios. No shading indicates that no capacity was deployed across any of the scenarios in that area.

There are fewer than 20 locations that see the highest levels of new capacity consistently across scenarios ($>5,000$ MW of capacity on average and sitings in all four scenarios; dark purple coloring). One of these locations is driven by natural gas deployment and can be seen in Figure 6 (upper right panel) near the southeast Washington border. Other high-intensity locations are due to onshore wind deployment (Figure 6, lower left panel). These are mostly in the eastern region of the Western Interconnection where wind speeds are strongest (e.g., Wyoming and

Colorado). High deployment intensity due to solar PV (Figure 6, lower right panel) is most common in the southern California region. Based on this analysis, these regions are projected to be particularly economically viable locations for new energy infrastructure in the future.

At a somewhat lower siting intensity level, a considerable number of areas see up to 5000 MW of capacity in all four scenarios. Wind and solar PV are the greatest contributors to these areas, particularly in the state of California and in southeastern Washington which have large clusters of consistent deployment. For non-renewables, there are a greater number of sporadic deployments of large capacity where >5000 MW is typically deployed but under only one or two scenarios. This is in part due to total capacity deployment differences in the capacity expansion model under different scenarios. For example, under the rcp45 scenarios there are fewer natural gas-fired facilities sited, meaning that there are locations that only see new sitings under the rcp85 scenarios. This further emphasizes the scenario-dependent community-level impacts demonstrated in Figure 5.

We also identify areas that never see any new power plant development at all (no coloring). These locations are either unsuitable for siting or are suitable but never saw any sitings across all scenarios as CERF's economic algorithm identified locations with lower net locational costs elsewhere. These tend to be more remote regions, including much of Nevada, central Montana, and New Mexico. Most of the siting in these states takes place closer to population areas that offer a greater number of grid interconnection locations and lower interconnection costs.

3.4. Feature Analysis of Economic Siting Drivers

The CERF power plant siting model considers multiple economic drivers in its siting decision-making. Two of these drivers vary spatially across suitable 1 km² locations: (1) the locational marginal value of new generation and (2) the cost of interconnecting to the electric grid. The tradeoff between these two factors drives siting towards areas that offer more valuable combinations of LMP and IC through the net locational cost. By analyzing the IC and LMP values for locations that received sitings and for those that did not, we can infer the relative influence of the two factors on CERF's decision-making for different technologies and determine how our scenarios influence those drivers. Figure 7 shows the results of our ML RF classifier with SHAP feature analysis for IC and LMP split out by scenario and technology type (see Methods). The value of each marker represents the mean of the absolute value of each individual SHAP value and indicates the level of influence that the specific feature (IC or LMP) has on whether a particular location saw a new power plant of the given technology type. To see individual SHAP feature values by technology and scenario, refer to Supplemental Figures 6-12.

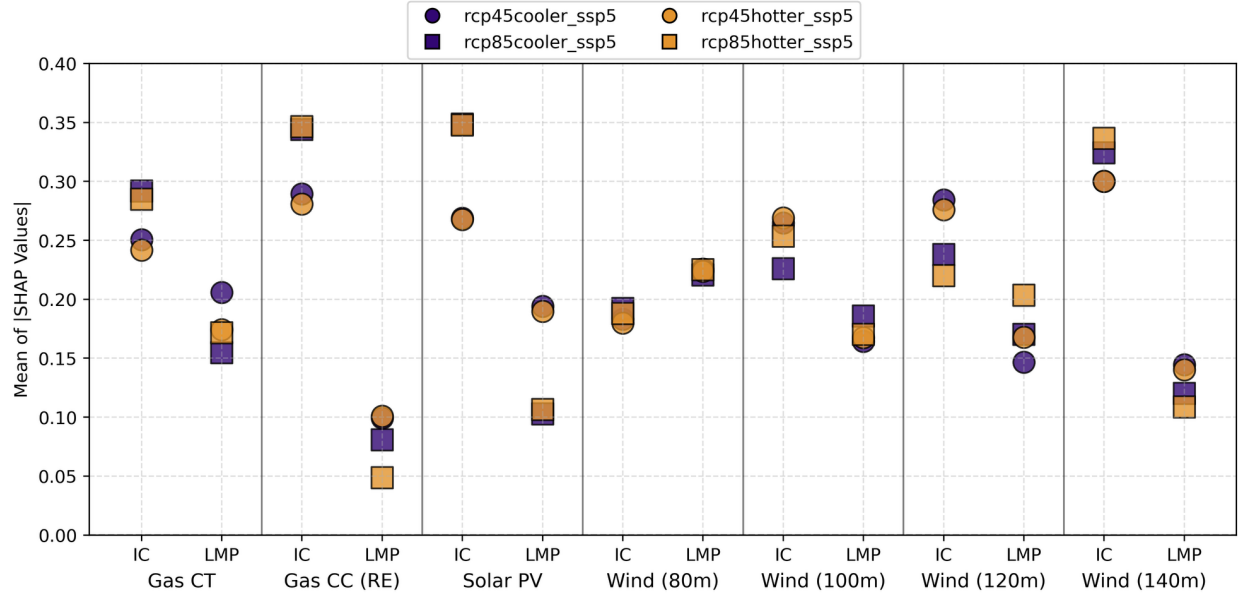


Figure 7 SHAP feature analysis results showing the mean of the absolute value of SHAP values by technology type and scenario for interconnection cost (IC) and locational marginal price (LMP). Values represent the relative feature importance in whether a specific siting location develops a new power plant. Generation technologies shown include natural gas combustion turbine (Gas CT), natural gas combined cycle with recirculating cooling (Gas CC [RE]) and no carbon capture sequestration, solar photovoltaic (solar PV), and onshore wind at 80m, 100m, 120m, and 140m hub heights.

Starting with technology type differences, Figure 7 shows that for all but one technology (Wind 80m), IC has greater feature importance than LMP for siting. Natural gas CC, solar PV, and onshore wind with higher hub heights show some of the largest spreads between IC and LMP feature importance. These results reflect the impact of the interaction between geospatial constraints and nearest interconnection points. Suitable natural gas CC locations are substantially constrained by cooling water proximity requirements, solar PV is constrained to areas with higher irradiance levels, and 140m wind faces both wind resource availability constraints and greater height-based siting restrictions, meaning that the locations that are suitable for siting may ultimately be far from grid interconnection points, making IC more influential. On the other hand, natural gas combustion turbines (CT) and wind with lower hub heights have comparatively more suitable land available for siting, which provides greater opportunity to site closer to substations. This makes IC less influential in determining the net locational cost as operational value becomes a larger factor in the economic decision-making.

For wind, there is an additional effect at play that is illuminated in Figure 7 due to the assumed hub height increase over time (growing from 80m in earlier years to 140m in later years), showing the temporal change in feature influence for the technology. Being one of the most deployed technologies across scenarios (see Figure 3), as wind is sited year over year, fewer suitable wind siting locations near substations remain available. Prime interconnection locations have already been taken by wind turbines of lower hub heights (or by other technologies) in previous years. The impact of this suitability shrinkage over time is displayed in Figure 7 when looking across all four hub heights in sequence. For the 80m hub height, which is sited in earlier years, there is an abundance of suitable space available closer to substations, making IC less important, and siting choices are driven by differences

in operational value instead. Moving forward in time, however, hub heights are assumed to increase due to technological improvements (an exogenous assumption in our experiment), and the difference in relative importance shrinks as we see look at each of these hub heights which increase with time, and IC surpasses LMP for hub heights 100m or higher and continues to grow incrementally. This implies that technologies that face substantial siting suitability constraints or see large amounts of deployment over time (e.g., wind) will have their siting decisions more heavily driven by interconnection availability if there is no additional interconnection infrastructure expansion.

Feature influence values are relatively consistent for each variable. Across all technologies, IC feature influence lies predominantly between 0.20 and 0.35 and for LMP it lies between 0.05 and 0.22, showing that there is some consistency in feature importance across technologies. Moreover, across our results, we find that there is only a small difference (generally within +/- 10%) in feature importance when comparing weather/emissions versus SSP influences. The rcp45 scenarios produce larger LMP feature influence over the rcp85 scenarios for all technologies other than wind. For natural gas CT and solar PV, the higher LMP influence is explained by higher operational values under the rcp45 scenarios. Overall, emission-and weather-based influences on siting appear to be mild once filtered through the larger influences of technology-specific siting restrictions and interconnection point availability. Figure SI 5 shows similar results for the SSP3 scenarios. While the choice of rcp and GCM sensitivity scenario appears to nudge the influence of IC or LMP on siting, Figure 7 shows that points are still clustered towards a common value for each technology type. This implies that the total amount of land both suitable and available for siting and the quantity of available interconnection points to the grid are ultimately the defining factors for whether a specific location sees development.

4. Limitations and Areas of Future Research

The CERF power plant siting model currently considers multiple economic factors in its net locational cost algorithm: interconnection costs, operating costs, and operating benefits. Additional economic factors that are not currently captured include land acquisition costs, insurance costs, environmental study costs, opportunity costs from land use tradeoffs, and the potential value from co-locating with other services such as agrivoltaics. Including these may guide CERF siting decisions towards new areas. Model enhancements in this direction could lead to improved projections of future buildouts.

Non-economic influences on siting are captured through the geospatial suitability constraints in CERF. While the model has hundreds of technology-specific geospatial layers representing a diversity of siting issues, all but population density are currently static and independent of the future scenario. The addition of scenario-dependent water availability for thermoelectric cooling, dynamic land use projections to exclude agricultural areas as they change over time, and dynamic hazard projections (e.g., flooding, drought, wildfires) are just a few examples of spatiotemporal aspects that could influence model outcomes. Expanding the number of dynamic spatial features and siting drivers being captured in our coupled modeling process may ultimately demonstrate new siting patterns and a greater variability in IC and LMP feature influence on siting across scenarios. Computational limitations stand as a current barrier to implementation, but future research will investigate methods to integrate additional models into the experiment framework so that we are able to capture additional scenario influences.

While our future scenarios capture uncertainty across a wide but plausible range of future weather-emissions constraint and socioeconomic conditions, our experiment does not fully characterize the uncertainty that appears across the integrated modeling chain. Assumptions and uncertainty introduced at each stage of the experiment may compound as they are passed from the top level (i.e., our weather and socioeconomic assumptions) through the capacity expansion model and ultimately into power plant siting model. For example, the coarse spatial resolution of GCAM-USA introduces uncertainty in the total capacity and types of technologies deployed in each state. This is then passed to CERF which identifies specific locations for plants, those specific locations and their interconnection points, in turn, impact grid operations and LMPs which impact future siting. Quantification of the uncertainty passed between, and possibly amplified by, our models is not yet well understood. Future research will conduct formal uncertainty characterization and sensitivity analyses across the modeling chain to identify how and where uncertainty is introduced and how that influences downstream model outcomes. This includes evaluating the sensitivity of siting results to capacity expansion assumptions, geospatial suitability, and economic parameters to better isolate the contribution of each model component on overall results.

5. Conclusions

The electric grid is facing numerous growing challenges from increasingly uncertain peak energy demand, the retirement of dispatchable resources, and the disruptive nature of extreme weather events. Understanding where and when different types of power plant infrastructure may be interconnected will be instrumental in developing robust planning and operational strategies for the future. To achieve this, new modeling frameworks that capture land, energy, water, and human dynamics concurrently is necessary.

Here, we present the results of our integrated multisector, multiscale modeling framework that projects the evolution of the electricity infrastructure landscape of the US Western Interconnection between 2020-2055 across a wide yet plausible range of futures. We model eight scenarios representing combinations of different weather conditions, emissions constraints, degrees of electrification, economic growth, technology growth, and population growth. We show that there are distinct scenario-driven differences in when and where the power plant landscape develops over time. These differences are predominantly driven by the quantity and type of generation capacity deployed in the capacity expansion plan and the associated amount of available land for siting in low interconnection cost areas. Specifically, we find that scenarios with the higher degree of electrification and emissions constraints lead to substantial amounts of solar PV and onshore wind, consistently built in southern California and eastern states of the interconnection, respectively, but with large clusters in other states such as Montana. Our scenarios without emissions constraints have less aggregate capacity by 2055, and a greater share of that capacity consists of natural gas-fired generation. These differences in the future technology resource mix dramatically shape the power plant landscape at local scales and imply considerable changes in land use. Our results also show that there are regions of the US Western Interconnection that are projected to see large amounts of development across all scenarios, primarily in southern California for solar, southeastern Washington for natural gas, and in areas in the east with high resource potential for onshore wind.

We also find that the relative influence of operational value (i.e., LMP) and interconnection costs (i.e., IC) on siting location varies by technology type. IC is found to be the most influential factor due to suitability-induced land constraints or because of large amounts of land competition due to large amounts of deployment. This finding implies that transmission expansion efforts that reduce IC will increase the relative influence of locational operating value on siting locations (if the transmission expansion does not significantly impact LMPs). This shift may be particularly true for technologies with relatively less suitable land, such as thermoelectric technologies that require adequate cooling water resources. Future research will explore the sensitivity of lower interconnection costs on LMP feature influence and examine how the power plant landscape may differ as a result.

Acknowledgments

This research was supported by the U.S. Department of Energy, Office of Science, as part of research in the MultiSector Dynamics, Earth and Environmental Systems Modeling Program. Pacific Northwest National Laboratory is a multi-program national laboratory operated by Battelle for the U.S. Department of Energy under Contract DE-AC05-76RL01830. This paper leverages the capabilities of the MultiSector Dynamics – Living, Intuitive, Value-adding, Environment (MSD-LIVE) platform funded by U.S. Department of Energy, Office of Science. MSD-LIVE, <https://msdlive.org>, is developed and maintained by Pacific Northwest National Laboratory.

Open Research

All the data and code supporting this paper is openly available. The data can be downloaded from Mongird (2025). The code is available at https://github.com/IMMM-SFA/mongird-et al_2026 under a BSD2-clause license (Mongird, 2026).

Conflict of Interest

The authors declare there are no conflicts of interest for this manuscript.

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